Database Management System (DBMS)

Previous Year Question Paper – 2017
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. What are ACID properties? Explain each with a short example. *(5 marks each)*

Q2. Define primary key and candidate key. Explain the difference between them. *(5 marks each)*

Q3. What is a foreign key? How does it help enforce referential integrity? *(5 marks each)*

Q4. Define a database transaction. Draw and explain the transaction state diagram. *(5 marks each)*

Q5. What is Boyce-Codd Normal Form (BCNF)? How is it different from 3NF? *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. What are DDL, DML and DCL in SQL? Give two examples of each. *(10 marks each)*

Q2. What is a JOIN in SQL? Explain INNER JOIN and OUTER JOIN with examples. *(10 marks each)*

Q3. Explain the concept of indexing and describe the structure of a B+ Tree index. *(10 marks each)*

Q4. What is a deadlock? Explain any two techniques used to prevent deadlocks in DBMS. *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. Take an unnormalized relation and normalize it step by step up to BCNF, explaining each step. *(15 marks each)*

Q2. Describe the phases of query processing in DBMS, including parsing, optimization and execution. *(15 marks each)*

Q3. Design a database for a University Management System and write SQL queries to retrieve, insert, update and delete records. *(15 marks each)*

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